

CellRank 2: Unified Fate Mapping in Multiview Single-cell Data

01. Paper Information

Paper	CellRank 2: unified fate mapping in multiview single-cell data
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02. At a Glance

No.	Aspect	Reading note
01	Question	How can we infer probabilistic cell-fate trajectories from diverse single-cell data views, especially when RNA velocity is unreliable or when time points, developmental potential, or metabolic labeling provide better directional information?
02	Input	Multiview single-cell data, including gene expression, RNA velocity, pseudotime, CytoTRACE developmental potential, experimental time points, and metabolically labeled RNA measurements.
03	Model	CellRank 2 separates transition-matrix construction from downstream trajectory analysis. Modality-specific kernels build Markov chains, and view-independent estimators such as GPCCA infer initial states, terminal states, fate probabilities, gene trends, and lineage-correlated genes.
04	Output	Initial, intermediate, and terminal states; fate probabilities for each cell; lineage-specific gene-expression dynamics; lineage-correlated genes; random-walk and circular visualizations; and kinetic-rate-informed regulatory programs.
05	Main claim	A modular kernel-estimator framework can unify fate mapping across RNA velocity, pseudotime, developmental potential, time-course data, and metabolic-labeling data while scaling to large single-cell datasets.
06	Why it matters	The key conceptual advance is to turn CellRank from an RNA-velocity-centered method into a general probabilistic framework for multiview single-cell dynamics. This makes fate mapping usable in many biological systems where one directional signal alone is insufficient.

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1 One-sentence summary

Core summary

CellRank 2 generalizes CellRank into a modular and scalable framework in which different biological data views define cell-cell transition kernels, while shared estimators analyze the resulting Markov chain to infer fate probabilities, terminal states, and lineage-specific gene programs.

2 Why this paper matters

The original CellRank solved an important problem: it transformed local RNA velocity information into global, probabilistic fate maps. However, RNA velocity is not always reliable. Some biological systems violate RNA velocity assumptions, some datasets do not contain sufficient spliced/unspliced signal, and some experimental designs provide more direct information through time points, developmental potential, or metabolic labeling.

CellRank 2 matters because it reframes fate mapping as a two-stage problem. First, a **kernel** converts one or more data views into a cell-cell transition matrix. Second, an **estimator** analyzes that transition matrix to infer biological trajectories. This decoupling is the main methodological move. Once a transition matrix is available, the same downstream machinery can be used for terminal-state discovery, fate probability estimation, gene-trend analysis, and driver-gene ranking.

Key takeaway

CellRank 2 is less a single new trajectory algorithm and more a unified framework: different sources of directionality can be plugged in as kernels, while the same probabilistic Markov-chain analysis produces interpretable fate maps.

3 Biological and computational background

3.1 The biological problem

Cells change state during development, regeneration, reprogramming, and disease progression. Single-cell RNA-seq captures many cells along these processes, but each cell is usually measured only once. Therefore, the computational task is to reconstruct a dynamic process from static or partially time-resolved observations.

Earlier methods often focused on one view of the data:

- **Pseudotime methods** order cells along a trajectory but often require a known initial state.
- **RNA velocity methods** infer local direction from spliced and unspliced transcripts but may fail when kinetic assumptions are violated.
- **Optimal transport methods** map cell populations across experimental time points but often ignore useful within-time-point variation.
- **Metabolic-labeling methods** provide time-resolved information about newly synthesized RNA but require specialized modeling of transcription and degradation.

Why CellRank 2 is needed

Problem 3.1

No single data view is universally reliable. A good fate-mapping framework should be able to use whichever source of directionality is most appropriate for the biological system: pseudotime when an initial state is known, developmental potential when it is unknown, experimental time when samples are collected across stages, or metabolic labeling when newly transcribed RNA is measured.

3.2 The computational framing

CellRank 2 retains the probabilistic Markov-chain view of CellRank. Each cell is a state, and edges encode transition probabilities from one cell to putative descendants. The key difference is that CellRank 2 no longer requires RNA velocity as the only source of directionality.

Formally, each kernel defines a transition matrix:

$$T \in \mathbb{R}^{n \times n}, \quad T_{ij} = P(x_i \rightarrow x_j), \quad (1)$$

where n is the number of cells and T_{ij} is the probability that cell i transitions toward cell j . After a kernel constructs T , an estimator analyzes the same object regardless of whether T came from RNA velocity, pseudotime, CytoTRACE, experimental time, or metabolic labeling.

Method note

The central abstraction is the transition matrix. Kernels answer: “How should we estimate cell–cell transitions from this data view?” Estimators answer: “What biological trajectories are implied by this transition matrix?”

4 Method breakdown

4.1 Step 1: build modality-specific kernels

CellRank 2 introduces a modular set of kernels. Each kernel converts a particular source of biological information into a directed transition matrix.

- **VelocityKernel:** uses RNA velocity, inherited from the original CellRank framework.
- **PseudotimeKernel:** biases a nearest-neighbor graph toward increasing pseudotime.
- **CytoTRACEKernel:** uses developmental potential or maturity estimates to orient transitions from naive to mature states.
- **RealTimeKernel:** combines inter-time-point transitions with intra-time-point similarity to model time-course data.
- **Metabolic-labeling-based velocity:** estimates cell- and gene-specific kinetic rates from labeled RNA and uses the resulting vector field for transition inference.

These kernels may also be combined. For example, two kernels $k^{(1)}$ and $k^{(2)}$ can be merged with a global weight α :

$$T = \alpha T^{(1)} + (1 - \alpha) T^{(2)}, \quad \alpha \in [0, 1]. \quad (2)$$

This enables multiview modeling when multiple directional signals are available.

4.2 Step 2: analyze the transition matrix with estimators

Once a transition matrix has been constructed, CellRank 2 uses view-independent estimators. The main estimator is based on generalized Perron cluster analysis (GPCCA), which coarse-grains the cell-level Markov chain into macrostates.

The macrostates are classified as:

- **Initial states:** populations that serve as likely starting points of trajectories.
- **Terminal states:** stable populations that cells are likely to reach and remain in.
- **Intermediate states:** transitional populations connecting initial and terminal states.

4.3 Step 3: compute fate probabilities

After terminal states are identified, CellRank 2 computes the probability that each cell eventually reaches each terminal state. This is the same conceptual absorption-probability problem as in CellRank, but CellRank 2 improves scalability by exploiting sparsity and by collapsing representative cells of a terminal state into a pseudo-state.

A useful abstraction is:

$$F_{ik} = P(\text{cell } i \text{ eventually reaches terminal state } k). \quad (3)$$

The resulting fate matrix F is used for visualization, lineage assignment, and downstream gene analysis.

Key takeaway

Fate probability is the common language across all kernels. Whether directionality comes from pseudotime, CytoTRACE, time-course mapping, RNA velocity, or metabolic labeling, the output is still a probabilistic map from current cells to future fates.

4.4 Step 4: identify lineage-correlated genes

CellRank 2 ranks genes by correlating gene expression with fate probabilities. For a gene g and terminal fate k , the basic idea is:

$$r_{g,k} = \text{corr}(x_{1g}, x_{2g}, \dots, x_{ng}; F_{1k}, F_{2k}, \dots, F_{nk}). \quad (4)$$

Genes with high positive correlation are interpreted as lineage-correlated genes or putative drivers of fate commitment. These genes can then be plotted as smooth expression trends along pseudotime or other continuous ordering variables.

4.5 Step 5: infer kinetic regulation from metabolic labeling

For metabolic-labeling data, CellRank 2 introduces an additional modeling step. It estimates cell- and gene-specific transcription and degradation rates from labeled RNA dynamics. These rates are then used to infer a velocity-like vector field and downstream transition probabilities.

For gene g in cell i , the conceptual model is:

$$\frac{dm_{ig}(t)}{dt} = \alpha_{ig} - \gamma_{ig}m_{ig}(t), \quad (5)$$

where α_{ig} is a transcription rate and γ_{ig} is a degradation rate. This enables analysis of whether lineage-associated genes are regulated through increased transcription, decreased degradation, or both.

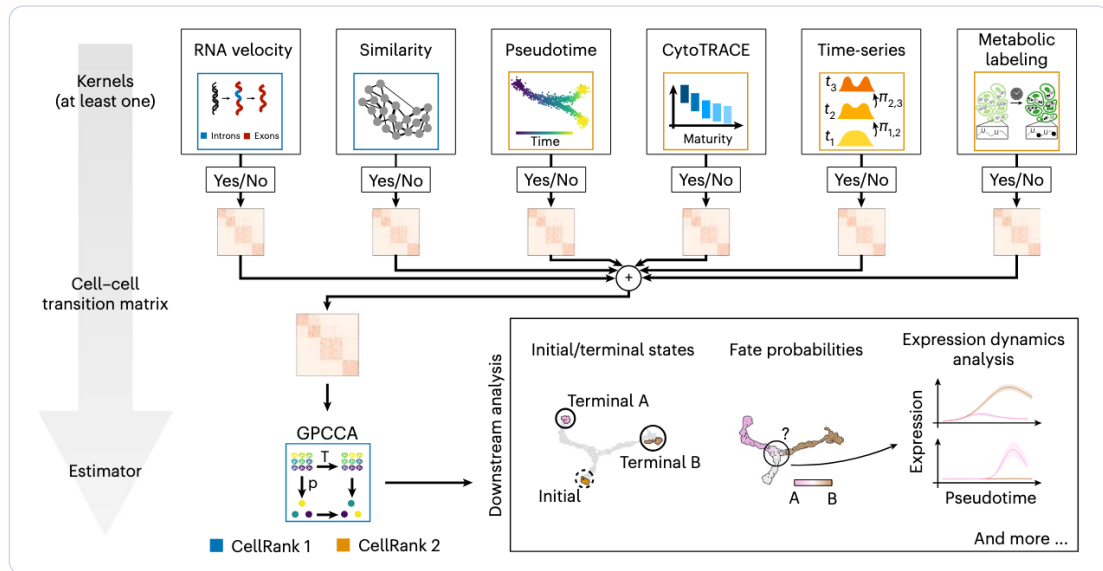


Figure 1: Main text Fig. 1. Overview of CellRank 2. This figure is the best conceptual summary of the kernel-estimator design and the extension from CellRank 1 to multiview fate mapping.

How to read Fig. 1

Figure note 4.1

The left side shows how different biological priors are converted into kernels. The middle transition matrix is the shared mathematical object. The right side shows that the same estimator module produces initial and terminal states, fate probabilities, and expression dynamics regardless of which kernel generated the transition matrix.

5 Main experiments and evidence

5.1 PseudotimeKernel in human hematopoiesis

The first application addresses a key limitation of RNA velocity. In human hematopoiesis, conventional RNA velocity can infer incorrect dynamics because its assumptions are violated. CellRank 2 therefore uses a PseudotimeKernel that biases transitions toward increasing diffusion pseudotime.

The PseudotimeKernel correctly recovers terminal states and the initial state in human hematopoiesis. It also produces a vector-field-like visualization and identifies lineage-associated transcription factors such as *TCF4* and *RUNX2* for the plasmacytoid dendritic cell lineage.

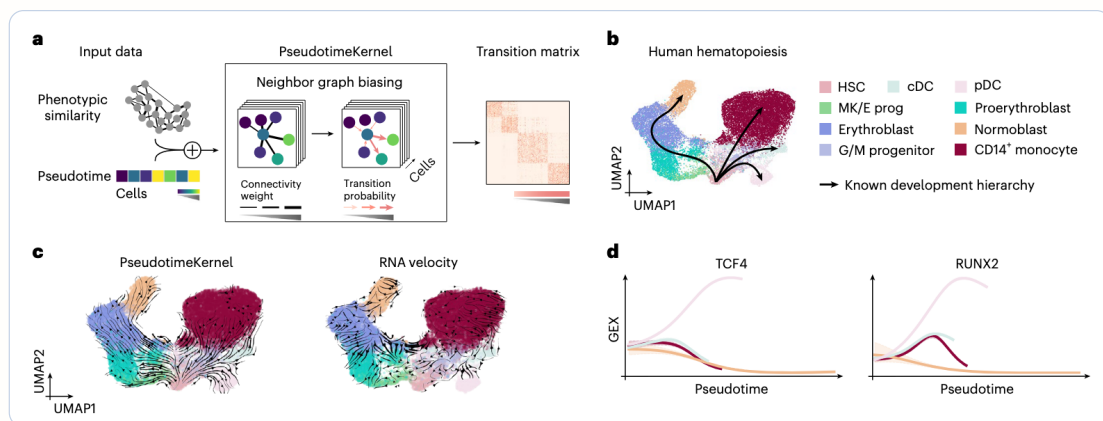


Figure 2: Main text Fig. 2a–d. Pseudotime-informed fate mapping in human hematopoiesis. CellRank 2 uses pseudotime to overcome RNA velocity failures in a system where velocity assumptions are violated.

Interpretation

Figure note 5.1

This experiment demonstrates that CellRank 2 is not tied to RNA velocity. When velocity is misleading, a biologically meaningful pseudotime can provide a better directional prior for fate mapping.

5.2 CytoTRACEKernel in embryoid body endoderm development

The second application uses developmental potential. Pseudotime methods usually require an initial state, but the initial state may not always be known. CytoTRACE provides a stemness or developmental-potential score, which CellRank 2 converts into directed transition probabilities through the CytoTRACEKernel.

In embryoid body development, the CytoTRACEKernel recovers most terminal populations and the correct initial state. It also identifies known endoderm-associated transcription factors, including *MIXL1*, *FOXA2* and *SOX17*. By plotting lineage-correlated genes along CytoTRACE pseudotime, the paper reconstructs a temporal activation pattern for endoderm development.

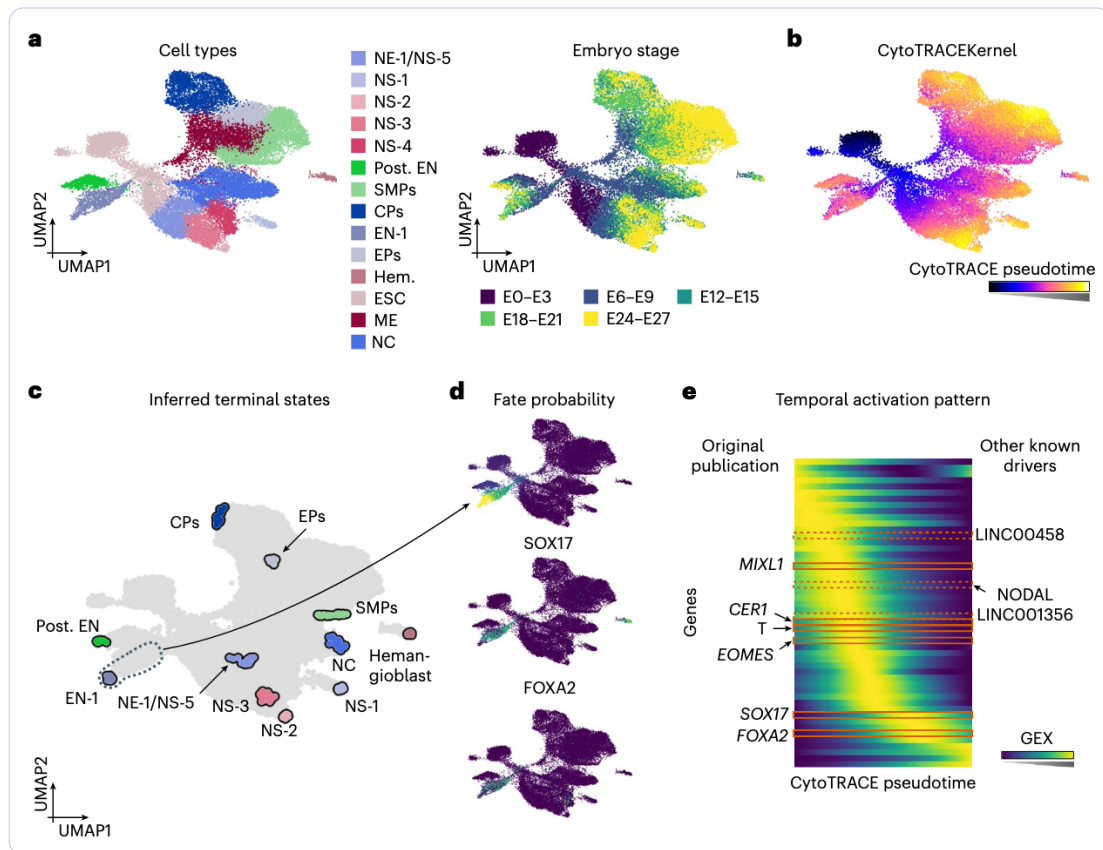


Figure 3: Main text Fig. 3a–e. CytoTRACEKernel analysis of embryoid body endoderm development. CellRank 2 uses developmental potential to recover terminal states, lineage drivers and temporal activation patterns without requiring a manually specified initial state.

Why this result matters

Figure note 5.2

This result shows how CellRank 2 can use a developmental-potential prior when pseudotime initialization is ambiguous. The key biological output is not just terminal-state detection, but also the ordering of lineage-correlated gene programs along developmental progression.

5.3 RealTimeKernel in pharyngeal endoderm development

The third application uses experimental time points. Standard optimal transport maps cells across consecutive time points but may ignore transitions within each time point. CellRank 2 introduces the RealTimeKernel, which combines inter-time-point transitions from Waddington optimal transport with intra-time-point transitions based on gene-expression similarity.

In pharyngeal endoderm development, the RealTimeKernel identifies most terminal states and improves terminal-state identification compared with an RNA-velocity-based analysis. A particularly important result is the discovery of a putative progenitor cluster with increased probability of becoming medullary thymic epithelial cells (mTECs). CellRank 2 also identifies relevant mTEC-associated genes and transcription factors such as *Fos*, *Grhl3* and *Elf5*.

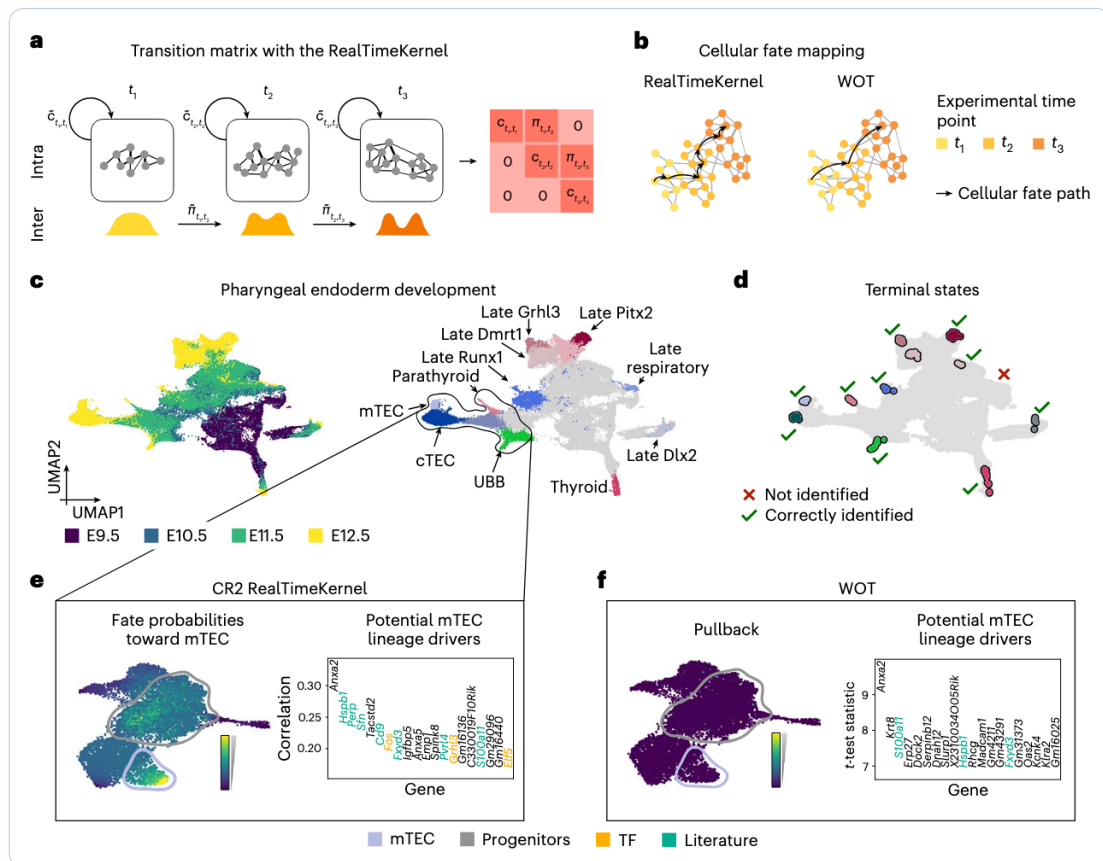


Figure 4: Main text Fig. 4a–f. RealTimeKernel analysis of pharyngeal endoderm development. By combining transitions within and across time points, CellRank 2 identifies mTEC fate probabilities and putative mTEC lineage drivers missed by WOT alone.

Interpretation

Figure note 5.3

The RealTimeKernel is the clearest example of multiview thinking in this paper. Across-time-point information gives temporal direction, while within-time-point similarity preserves local developmental structure. Their combination produces a more continuous and biologically informative fate map.

5.4 Metabolic labeling in intestinal organoids

The final main application uses metabolic-labeling data. Instead of relying on spliced/unspliced RNA velocity, CellRank 2 estimates transcription and degradation rates from labeled RNA dynamics in pulse–chase experiments. These kinetic rates are used to infer transition probabilities and fate decisions.

In murine intestinal organoids, this approach recovers terminal states corresponding to enterocyte, enteroendocrine, goblet and paneth lineages. Compared with dynamo, CellRank 2 identifies more terminal states. Compared with CellRank 1 using classical RNA velocity, CellRank 2 improves terminal-state purity and driver-gene ranking. The estimated kinetic rates further allow the authors to distinguish regulatory strategies, such as cooperative and destructive regulation, along goblet and enterocyte lineages.

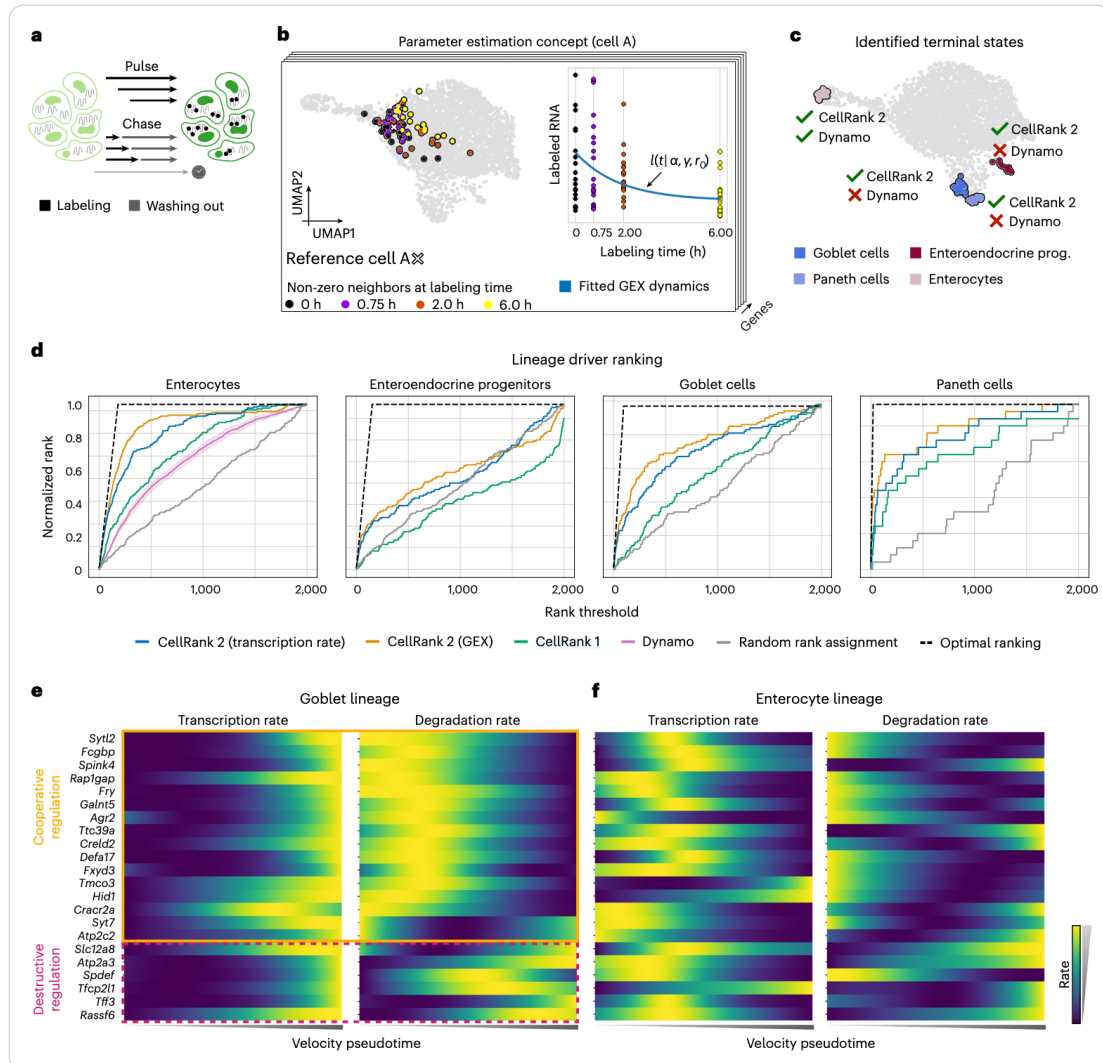


Figure 5: Main text Fig. 5a–f. Metabolic-labeling-based fate mapping. CellRank 2 estimates kinetic rates from labeled RNA, recovers intestinal organoid terminal states and links fate decisions to transcriptional and degradation-rate regulation.

Why this result is important

Figure note 5.4

This experiment extends fate mapping beyond transcript abundance and splicing ratios. By estimating transcription and degradation rates, CellRank 2 can ask not only which genes correlate with a fate, but also how those genes are dynamically regulated.

6 Core mathematical intuition

6.1 Why kernels?

A kernel is a rule for constructing cell–cell transition probabilities. The same cell population can be analyzed through different biological views, and each view may imply a different transition matrix. CellRank 2 formalizes this through kernels.

Different views, same target

Intuition 6.1

RNA velocity, pseudotime, CytoTRACE, experimental time and metabolic labeling all provide different clues about directionality. CellRank 2 converts each clue into a transition matrix so that the downstream fate analysis remains consistent.

6.2 Why estimators?

Estimators are view-independent. Once a transition matrix exists, it can be coarse-grained, analyzed for terminal states and used for absorption probabilities. This is why CellRank 2 can support new data modalities without rewriting the fate-probability machinery.

6.3 Why sparse Markov chains matter

Single-cell datasets can contain hundreds of thousands or millions of cells. A dense transition matrix would be computationally prohibitive. CellRank 2 assumes that each cell can transition only to a small set of possible descendants, producing sparse transition matrices and enabling scalable computation.

6.4 Why multiview modeling is useful

Different data views capture different temporal or biological scales. Experimental time points provide global temporal direction, but within-time-point similarity captures asynchronous local progression. Metabolic labeling gives short-timescale RNA dynamics, whereas pseudotime provides a global developmental ordering. Combining views can therefore yield a more faithful trajectory than relying on a single source.

7 Strengths of the paper

- **Clear conceptual generalization:** the paper turns CellRank from an RNA velocity method into a modular fate-mapping framework.
- **Strong software design:** separating kernels from estimators makes the framework extensible to new data modalities.
- **Multiple directional priors:** pseudotime, CytoTRACE, experimental time and metabolic labeling can all be used to construct transition matrices.
- **Scalability:** sparse transition matrices and improved fate-probability computation make the framework suitable for large datasets.
- **Biological breadth:** applications cover hematopoiesis, embryoid body differentiation, pharyngeal endoderm development and intestinal organoids.
- **Interpretability:** fate probabilities are linked to lineage-correlated genes, gene-expression trends and kinetic regulation strategies.
- **Practical value:** users can choose the kernel that best matches their dataset and biological assumptions.

8 Limitations and critical reading

Critical reading

CellRank 2 is flexible, but flexibility also shifts responsibility to the user. The quality of the result depends on whether the chosen kernel matches the biology and assumptions of the dataset.

8.1 Kernel choice is assumption dependent

Each kernel has its own assumptions. The PseudotimeKernel assumes that the pseudotime is meaningful and oriented correctly. The CytoTRACEKernel assumes that developmental potential decreases along differentiation. The RealTimeKernel assumes that experimental time points are informative and sufficiently close relative to the dynamics. The metabolic-labeling approach assumes that labeled RNA data can reliably support kinetic-rate estimation.

8.2 Correlation does not prove causality

CellRank 2 ranks putative driver genes by correlating gene expression with fate probability. This is useful for hypothesis generation, but correlation alone cannot prove that a gene causally drives a fate decision. Perturbation data would be needed for stronger causal claims.

8.3 Multiview combination weights are not automatically biological

Kernel combinations can integrate multiple data views, but global weights may not reflect local biological heterogeneity. A cell near one region of the manifold might need a different weighting of pseudotime, velocity or experimental time than a cell elsewhere.

8.4 Metabolic labeling requires specialized data

The kinetic-rate analysis is powerful but requires pulse–chase or related metabolic-labeling protocols. This limits immediate applicability to datasets where such measurements are available.

9 What I can reuse for my own research

Implications for my work

The most reusable idea is the kernel-estimator separation. For pathology-omics modeling, different biological or clinical views could define different kernels, while a shared probabilistic estimator could infer disease-state transitions, phenotype probabilities or treatment-response trajectories.

Potential reusable ideas:

- **Kernel design:** define a transition matrix from each modality, such as pathology morphology, spatial omics, single-cell expression, perturbation response or clinical time.
- **Multiview transition modeling:** combine kernels to integrate complementary biological evidence.
- **Probabilistic endpoint prediction:** replace hard state assignment with probabilities toward disease phenotypes or therapeutic response endpoints.
- **Driver discovery:** correlate endpoint probabilities with genes, pathways or tissue features to identify candidate regulators.
- **Kinetic interpretation:** when temporal or perturbation data are available, distinguish whether phenotype-associated genes are driven by activation, repression or degradation-like processes.

10 Possible extensions

10.1 Perturbation-aware CellRank 2

Perturb-seq or CRISPR-screen data could define a perturbation kernel. This would allow transition probabilities to encode how gene perturbations move cells toward or away from specific terminal phenotypes.

10.2 Spatial CellRank 2

Spatial omics could contribute a spatial-neighborhood kernel. This would be useful for modeling tissue-level fate decisions, tumor invasion fronts, immune exclusion or spatially organized differentiation.

10.3 Pathology-omics adaptation

In computational pathology, cells, patches or tissue regions could be nodes in a graph. Histology morphology, gene expression, spatial context and clinical time could define separate kernels. A CellRank-2-like estimator could then infer probabilistic trajectories toward progression, metastasis or therapy resistance.

11 Questions to ask while reading

Questions for further thought

1. How should one choose the most appropriate kernel for a new dataset?
2. When multiple kernels disagree, should we trust one view or combine them?
3. How sensitive are terminal states and fate probabilities to kernel weights?
4. Can local, cell-dependent kernel weights improve multiview integration?
5. Which lineage-correlated genes remain robust across different kernels?
6. How can perturbation experiments validate driver genes inferred only by correlation?

12 My final evaluation

Table 1: Personal evaluation of the paper.

Criterion	Evaluation
Novelty	High. The paper generalizes CellRank into a modular framework that supports multiple data views rather than depending only on RNA velocity.
Biological value	High. The applications show that different kernels can recover meaningful fate maps in hematopoiesis, endoderm development, pharyngeal development and intestinal organoids.
Technical clarity	Strong. The kernel-estimator separation is simple, reusable and easy to explain.
Validation	Strong overall. The paper uses diverse datasets and compares against RNA velocity, WOT, dynamo and other relevant alternatives.
Main caveat	The framework is only as reliable as the chosen kernel and its assumptions. Driver-gene analysis is still correlational rather than causal.
Reading rating	★ 4.8/5

Key takeaway

CellRank 2 is best understood as a general language for probabilistic fate mapping. Its main contribution is not a single kernel, but the separation between how transitions are built and how biological trajectories are interpreted.